

E03 Equilibrium limited PFR example

In the lecture on membrane reactors, we solved the following example for both a membrane reactor and a PFR: The elementary reaction $A \rightleftharpoons H_2 + C$ is carried out isothermally and isobarically in a membrane reactor. The feed is pure A, with $C_{A0} = 0.2$ M and $F_{A0} = 10$ mol/min. The forward rate constant $k_f = 0.7 \text{ min}^{-1}$ and the equilibrium constant $K_c = 0.05$ M. Only H_2 can diffuse through the membrane. The mass transfer coefficient $k_c = 0.2 \text{ min}^{-1}$. The membrane reactor has a volume of 500 L. Determine the conversion for this membrane reactor.

For review, let's solve the PFR part of the example again, using two methods:

- First, as a PFR, but using the balance in terms of X instead of F:

$$\frac{dX_A}{dV} = \frac{-r_A}{F_{A0}}, \quad -r_A = k_A f \left(C_A - \frac{C_C C_{H2}}{K_c} \right)$$

$$C_A = \frac{C_{A0}(1-X_A)}{1+\varepsilon X_A}, \quad \varepsilon = Y_{A0} \delta, \quad \delta = 1$$

$$C_A = \frac{C_{A0}(1-X_A)}{1+X_A}$$

$$C_{H2} = \frac{C_{A0} X_A}{1+X_A} = C_C$$

$$-r_A = k_A f \left[\frac{C_{A0}(1-X_A)}{1+X_A} - \frac{C_{A0}^2 X_A^2}{(1+X_A)^2 K_c} \right]$$

$$\frac{dX_A}{\left[\frac{C_{A0}(1-X_A)}{1+X_A} - \frac{C_{A0}^2 X_A^2}{(1+X_A)^2 K_c} \right]} = \frac{K_A F V}{F_{A0}}$$

need to integrate from $X_A = 0$ to $X_A = 0.447213...$

$X_e = 0.447213$ this is the max conversion possible, aka the X_A @ the reactor exit

- Now let's solve for the equilibrium conversion ... To do that, we don't need to solve any reactor design equations:

$$K_c = \frac{C_c C_{H_2}}{C_A} = 0.05 \text{ M}$$

$$0.05 = \frac{C_{A0}^2 X_e^2}{(1+X_e)^2} \cdot \frac{1+X_e}{C_{A0}(1-X_e)} = \frac{C_{A0} X_e^2}{(1+X_e)(1-X_e)}$$

$$\underline{X_e = 0.447213}$$